

LAVANYA TYAGI

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Jaypee Institute of Information Technology

B.Tech. in Computer Science Engineering

2024 – 2028

Noida, India

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks, Software Engineering

Mount Carmel School

Class XII: 97% | Class X: 98%

Ghaziabad, India

Experience

Unicure India Pvt. Ltd.

Software Engineer Intern

May 2026 – July 2026

- Developed the company's responsive corporate website using **React.js** and **Tailwind CSS**, implementing reusable components across desktop, tablet, and mobile devices.
- Contributed to **PharmaIntel AI**, a pharmaceutical ERP and Production Intelligence System using **React.js**, **Node.js**, **Express.js**, and **MySQL**, with modules for medicine management, inventory, production cost estimation, and analytics.
- Developed and integrated **REST APIs** between frontend and backend services, and incorporated **Google Gemini AI** for inventory optimization, production planning, and data-driven recommendations.
- Tested APIs using **Postman** and validated database operations using **MySQL Workbench** while debugging application functionality and ensuring data consistency.

Projects

User Authentication System

Java · JDBC · Oracle

- Developed a multi-screen authentication system supporting registration, login, password recovery, and password change.
- Integrated **Oracle Database** with **JDBC** for credential storage, validation, and user data management.

AI Music Recommendation System

Python · Flask · Machine Learning

- Developed an AI-powered music recommendation system using **Flask** and machine learning techniques.
- Implemented personalized recommendation logic using dataset filtering and designed modular Flask backend APIs.

Smart City Emergency System

C++ · Graphs · DSU

- Developed an emergency routing optimization system for smart-city response scenarios using graph algorithms.
- Applied **Disjoint Set Union (DSU)** for path optimization and implemented an emergency response simulation.

Real-Time Chat Application

React.js · Firebase

- Built a real-time messaging application with user authentication and Firebase integration.
- Developed a responsive **React.js** frontend and integrated Firebase real-time database functionality.

Technical Skills

Languages: Java, C++, Python, JavaScript, SQL

Frontend: HTML, CSS, React.js, Tailwind CSS

Backend: Node.js, JDBC, Java Servlets, JSP

Databases: MySQL, Oracle, Firebase

Concepts: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Machine Learning, REST APIs

Tools: Git, GitHub, Eclipse, IntelliJ IDEA, VS Code, Apache Tomcat, Postman

Certifications

- Fundamentals of Remote Sensing** – NASA ARSET
- Core Java** – Abdul Bari
- Advanced Java: JDBC, Servlets and JSP**